

A Comparison of Bone–Patellar Tendon–Bone and Bone–Hamstring Tendon–Bone Autografts for Anterior Cruciate Ligament Reconstruction

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Background: Most of the previous comparative studies between patellar tendon and hamstring tendon anterior cruciate ligament grafts compared grafts of different constructs fixed with different methods.

Purpose: To compare patellar tendon and hamstring tendon grafts with the same fixation method used to reconstruct the anterior cruciate ligament.

Study Design: Randomized controlled trial; Level of evidence, 1.

Methods: During the reconstructive procedure, the hamstring tendon graft was prepared as a bone–hamstring–bone graft; both bone–patellar tendon–bone and bone–hamstring–bone grafts were fixed with interference screws. Eighty consecutive patients who underwent anterior cruciate ligament reconstruction were randomly assigned to either bone–patellar tendon–bone or bone–hamstring–bone groups. Follow-up examinations were performed for at least 5 years postoperatively. Seventy-two of the 80 patients (37 patients in the bone–patellar tendon–bone group and 35 in the bone–hamstring–bone group) were evaluated, with a mean follow-up period of 87.0 and 80.8 months, respectively. Follow-up examinations were performed using the International Knee Documentation Committee knee ligament standard and subjective knee forms.

Results: The mean KT-1000 arthrometer evaluation results showed no significant difference between the bone–patellar tendon–bone and bone–hamstring–bone groups (1.2 ± 2.1 mm and 1.7 ± 1.4 mm, respectively; $P = .24$). However, symptoms related to graft harvest (anterior kneeling pain) were more frequently observed in the bone–patellar tendon–bone group, and unsatisfactory results were correlated with severe kneeling pain in 3 patients from this group ($P = .0056$). Significant hamstring muscle weakness without complaint of functional deficit was found in the bone–hamstring–bone group ($P = .0045$).

Conclusion: Bone–hamstring–bone grafts were shown to reduce the risk of problems at the graft harvest site compared to bone–patellar tendon–bone grafts, with comparable results in the remaining clinical parameters tested.

Keywords: anterior cruciate ligament (ACL); ligament reconstruction; patellar tendon; hamstring tendon

Bone–patellar tendon–bone (BTB) autografts are the most popular choice for ACL reconstruction, and satisfactory results have been reported. In addition, the recent use of multistrand hamstring tendon autografts has also increased in popularity,

and many comparative studies between BTB and hamstring tendon grafts have been reported.[§] Spindler et al,²⁴ in their meta-analysis, demonstrated that the BTB graft had a slightly better static knee stability compared with the hamstring tendon graft, whereas an increased incidence of postoperative anterior knee pain at the graft harvest site concomitant with BTB graft has been indicated by other investigators.^{11,24} In contrast, there have also been some reports demonstrating no significant difference between the 2 grafts.^{6,18}

These inconsistent results in the comparison of the 2 grafts may be partly due to the variability of the reconstructive

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TABLE 1
Patient Demographics^a

	BTB Group	BHB Group	P
Age, y	23.7 ± 7.7	24.4 ± 9.7	.74
Sex (no. of subjects), female/male	16/21	20/15	.24
Time from injury to surgery, mo	11.4 ± 9.9	8.9 ± 9.9	.29
Follow-up period, mo	87.0 ± 12.2	80.7 ± 13.2	.10

^aValues are given as means ± SDs unless otherwise indicated. BTB, bone–patellar tendon–bone graft; BHB, bone–hamstring tendon–bone graft.

procedure employed in the previous studies. First, the graft construct between the 2 grafts is different: the BTB graft has a bone plug at each end, whereas the hamstring tendon graft has generally been used as a multistrand free-tendon graft without a bony component. Regarding the fixation method, the BTB graft has most commonly been fixed in the bone tunnel using interference screws. In contrast, the hamstring tendon graft is usually fixed at an extra-articular position away from the original insertion using a variety of methods, presenting a theoretical disadvantage for this graft construct. Thus, most of the previous studies comparing the BTB and hamstring tendon grafts have faced the inherent problem of comparing 2 different grafts used in different procedures.

With the intent of overcoming the disadvantages associated with the use of the hamstring tendon graft, Morgan et al¹⁹ and Liu et al¹⁷ came up with the bone–hamstring tendon–bone (BHB) graft, in which plugs taken from bone tunnel sites were attached to both ends of the graft. The reconstructive procedure in the BHB graft, including graft construct and fixation, is comparable to that of the BTB graft in ACL reconstruction. However, no comparative study between the BTB and BHB grafts has been reported.

In our study, a prospective and randomized comparison of the BTB and BHB grafts in ACL reconstructions was conducted. The purpose of this research was to make a critical comparison between the patellar tendon and the hamstring tendon autografts used to reconstruct the ACL using almost the same procedure.

MATERIALS AND METHODS

Subjects

From April 1994 to March 1996, a total of 80 consecutive patients with symptomatic ACL insufficiency underwent arthroscopically assisted ACL reconstructions at our institution. A prospective, randomized comparison was performed, in which the ACL was reconstructed with either an ipsilateral BTB or BHB graft according to the patient's birth date. Inclusion criteria were unilateral insufficiency of the ACL, no previous ligament reconstruction of the affected knee, and no concomitant ligament injury of the affected knee except for a grade I or II injury of the medial collateral ligament. The study design was approved by the Institutional Review Board of the Kobe University Graduate School of Medicine, and all patients provided their informed consent.

TABLE 2
Cause of Injuries^a

	BTB Group	BHB Group
Sporting activity		
Basketball	8	7
Soccer	4	4
Volleyball	5	4
Skiing	11	5
Other sport	6	12
Other (accident)	3	3

^aBTB, bone–patellar tendon–bone graft; BHB, bone–hamstring tendon–bone graft.

Follow-up examinations were performed for at least 5 years postoperatively. Among the 80 patients included in the study, 8 were lost to follow-up. Of the remaining 72 patients, 37 patients (21 male, 16 female) were in the BTB group and 35 patients (15 male, 20 female) in the BHB group. The mean follow-up periods in the BTB and the BHB groups were 87.0 months (range, 60–102 months) and 80.8 months (range, 60–103 months), respectively. There were no significant differences between the 2 groups in gender distribution and follow-up period ($P = .24$ and $.10$, respectively). Patient demographic data for other parameters such as age at surgery and time from injury to surgery were comparable in both groups (Table 1).

In both groups, more than 90% of the subjects were evaluated with sports-related injuries; no significant difference was observed between the groups in terms of cause of injury (Table 2).

Surgical Technique

All patients underwent an arthroscopically assisted, single-incision ACL reconstruction by a single surgeon (M.K.). Routine diagnostic arthroscopy was performed before ACL reconstruction. For concomitant intra-articular injuries, additional procedures such as meniscal repair using the inside-out technique, partial meniscectomy, and removal of chondral fragment were performed when required. The reconstructive procedure was performed using a leg holder and a tourniquet.

Bone–Patellar Tendon–Bone Graft. At graft harvest, the central third of the patellar tendon with a 2.0-cm bone

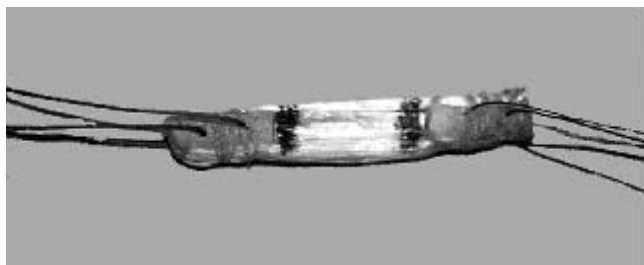


Figure 1. Bone-hamstring tendon-bone graft.

TABLE 3
Concomitant Meniscal Surgeries^a

	BTB Group	BHB Group
Meniscectomy		
Medial	7	7
Lateral	7	4
Bilateral	1	4
Meniscal repair		
Medial	11	8
Lateral	3	1
Bilateral	0	1
Repair/meniscectomy		
Medial/lateral	1	3
Lateral/medial	0	1

^aBTB, bone-patellar tendon-bone graft; BHB, bone-hamstring tendon-bone graft.

block at each end was taken. Each bone block at the end of the patellar tendon was trimmed, and two No. 1 absorbable sutures were passed through the drill holes. A tibial tunnel, 10 mm in diameter, was created at the center of the ACL attachment using an ACL endoscopic guide system (Smith & Nephew Inc, Andover, Mass). A femoral tunnel, 9 mm in diameter, was drilled through the tibial tunnel on the lateral wall of the femoral intercondylar notch at the position corresponding to 1 to 2 o'clock for the left knee (10 to 11 o'clock for the right knee) and 7 mm distal to the over-the-top position. The graft was inserted through the tibial tunnel, and femoral fixation was achieved with a 7 × 20-mm interference screw (DePuy, Warsaw, Ind). After the graft was secured on the femoral side, pretension was performed with a 50-N load applied for 1 minute. Then, the graft was fixed with a 9 × 20-mm interference screw (DePuy) while a tension of 20 to 30 N was maintained with the knee in full extension. Notchplasty was performed when necessary. The bony defects at the graft harvest site were filled with the bone meal from the tunnels and the bone pieces obtained during trimming of the harvested bone plug. The defect of the patellar tendon was superficially reapproximated with No. 1 absorbable sutures.

Bone-Hamstring Tendon-Bone Graft. Through a 4-cm longitudinal incision over the pes anserinus, the semitendinosus and gracilis tendons were harvested using a tendon stripper

(Mitek, Norwood, Mass) with the distal ends detached. The femoral and tibial tunnels were positioned in the same manner as the BTB grafts. In the BHB graft procedure, the tibial tunnel was created using a coring reamer (Arthrex, Naples, Fla), and the harvested bone plug was split in half longitudinally. Each of the bone blocks was trimmed to 20 mm in length, and 2 drill holes were created. The harvested semitendinosus and gracilis tendons were made into a 5-strand graft, 8 cm in length. After the 2 bone plugs were placed at both ends of the tendon graft, the tendon and bone plug were tied together in a circular fashion with No. 2 absorbable sutures. In addition, two No. 1 absorbable sutures were passed through the drill holes in each bone plug and used as stay sutures (Figure 1). The BHB graft was fixed to the femur and tibia in the same manner as the BTB graft, with the interference screws placed between the bone plug and the bone tunnel wall. During tibial fixation, as with the BTB graft procedure, a tension of 20 to 30 N was applied with the knee in full extension after pretensioning.

Postoperative Management

For all patients, the same rehabilitation protocol was applied. Immediately after surgery, the knee was immobilized at full extension using a hinged knee brace for 1 week. Intermittent passive motion exercise using a continuous passive motion machine was started from the first postoperative day. At 2 weeks, patients were allowed to walk with partial weightbearing, and closed kinetic chain exercise was started; full weightbearing was allowed at 4 weeks. A knee brace was worn for 8 weeks. Open kinetic chain exercise was allowed at 12 weeks. Jogging was permitted at 4 months, with gradual return to sports activity at 6 to 10 months.

Follow-up Evaluation

Follow-up examination was performed using the International Knee Documentation Committee (IKDC) knee ligament standard evaluation form¹² and the IKDC subjective knee form¹³; results were categorized using IKDC grades A (normal), B (nearly normal), C (abnormal), or D (severely abnormal). A side-to-side difference in KT-1000 arthrometer measurement at manual maximum stress was adopted as a parameter in the stability assessment, and heel height difference (HHD) was measured to evaluate the lack of extension. Isokinetic muscle strength testing was performed at 60 and 180 deg/s with a Cybex II dynamometer (Lumex Inc, Ronkonkoma, NY).

Statistical Analysis

Statistical analysis was performed with StatView software (SAS Institute, Cary, NC), and statistical power analysis was performed with G*Power, version 2.0.⁸ The Student *t* test was used to compare subjective assessment, HHD, KT-1000 arthrometer results, and muscle strength; the χ^2 test was used to compare the other patient data between the 2 groups. Significance was set at *P* < .05.

TABLE 4
Results of the IKDC Knee Ligament Standard Evaluation^a

	A (Normal)	B (Nearly Normal)	C (Abnormal)	D (Severely Abnormal)	P
I, Subjective assessment					
BTB group	25	10	2		
BHB group	26	8	1		.59
II, Symptoms					
BTB group	24	10	3		
BHB group	24	10	1		.33
III, Range of motion					
BTB group	30	7			
BHB group	32	3			
IV, Ligament examination					
BTB group	28	7	2		
BHB group	28	7			.16
V, Compartment findings					
BTB group	29	7	1		
BHB group	28	6	1		.97
VI, Harvest site morbidity					
BTB group	23	8	3	3	
BHB group	26	8	1		.056
VII, Radiograph findings					
BTB group	22	12	3		
BHB group	22	10	3		.94
VIII, Functional test results					
BTB group	24	10	3		
BHB group	25	9	1		.33
Final evaluation					
BTB group	11	16	7	3	
BHB group	9	21	5		.18

^aThe χ^2 test was used to compare grades A and B with grades C and D between the 2 groups. IKDC, International Knee Documentation Committee; BTB, bone–patellar tendon–bone graft; BHB, bone–hamstring tendon–bone graft.

RESULTS

Intraoperative Findings and Concomitant Surgeries

Of the 37 patients in the BTB group, 30 patients (81%) underwent surgery for meniscal tears during the ACL reconstruction, whereas 24 of the 35 patients (69%) in the BHB group underwent concomitant meniscal surgeries (Table 3). Although damage to the articular cartilage was observed in 14% of the patients in both groups, none required surgical intervention. There was no significant difference between the 2 groups in terms of the incidence of concomitant procedures for associated meniscal injuries ($P = .22$).

Activity Level

Before the injury, 92% (34/37 patients) of the BTB group and 89% (31/35 patients) of the BHB group participated in either strenuous or moderate sports activities (IKDC grades I or II¹²). Postoperatively, 28 patients (75.7%) in the BTB group and 24 patients (68.6%) in the BHB group returned to their preinjury activity levels. Nine patients in the BTB group and

11 in the BHB group decreased their activity levels. When comparing the return to preinjury activity levels, there was no significant difference between the 2 groups ($P = .16$).

IKDC Knee Ligament Standard Evaluation

Subjective Assessment. According to the IKDC subjective knee rating form,¹³ mean scores were 84.3 ± 11.0 for the BTB group and 86.8 ± 8.3 for the BHB group. No significant difference was demonstrated between the groups ($P = .38$) (Table 4).

Range of Motion. Range of motion was graded as A or B in patients from both groups, and almost all patients had full extension and flexion. An extension deficit of 5° was observed in 3 patients in the BTB group and 2 patients in the BHB group, and 1 patient had an extension deficit of 10° in the BTB group. A flexion deficit of 10° was observed in 2 patients in the BTB group and 1 patient in the BHB group. The mean HHD at final follow-up was 8.0 ± 14.1 mm in the BTB group and 6.9 ± 13.4 mm in the BHB group. Five patients in the BTB group and 3 in the BHB group had an HHD of greater than 20 mm. No patient exhibited

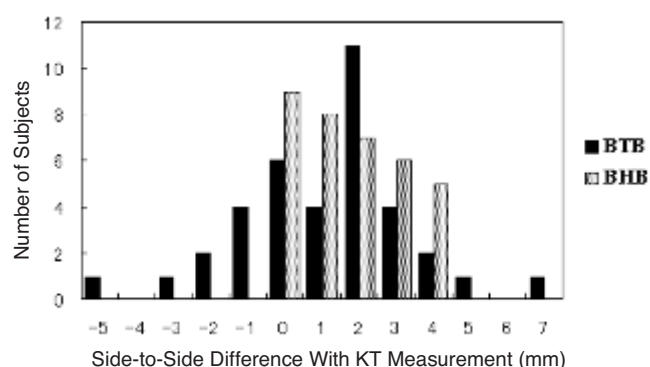


Figure 2. Results of KT-1000 arthrometer evaluation at follow-up.

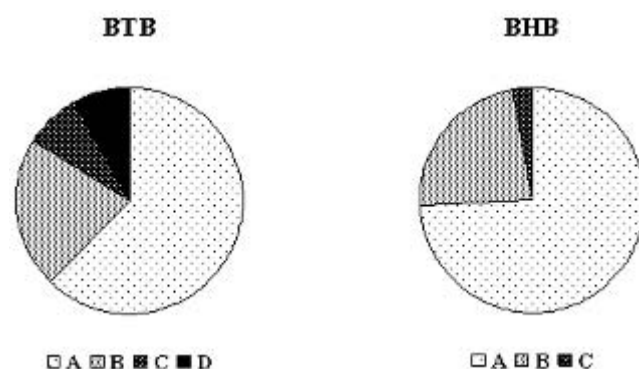


Figure 3. Results of harvest site morbidity at follow-up.

TABLE 5
Side-to-Side Comparison of Peak Torque Values^a

Isokinetic Test	BTB Group		BHB Group		P
	Mean	SD	Mean	SD	
Quadriceps					
60 deg/s	86.4	18.2	92.3	14.2	.26
180 deg/s	94.5	22.0	93.0	10.9	.79
Hamstring					
60 deg/s	100.0	16.6	85.6	13.1	.0045 ^b
180 deg/s	97.0	10.5	91.4	13.5	.14

^aBTB, bone–patellar tendon–bone graft; BHB, bone–hamstring tendon–bone graft.

^bSignificant difference between the 2 groups ($P < .05$).

an HHD of more than 50 mm. There were no significant differences between the groups in extension deficit, flexion deficit, or HHD ($P = .34$, $.58$, and $.96$, respectively).

Ligament Examination. The KT-1000 arthrometer evaluation at final follow-up showed that in the BTB group, the mean side-to-side difference was 1.2 ± 2.1 mm; a side-to-side difference of more than 3 mm was observed in 4 of 37 patients (10.8%). In the BHB group, the mean side-to-side difference was 1.7 ± 1.4 mm; 5 of 35 patients (14.3%) had greater than 3 mm side-to-side difference at final follow-up (Figure 2). There was no significant difference between the groups in this parameter ($P = .24$, power = 0.71).

Harvest Site Morbidity. Of the 37 patients in the BTB group, symptoms related to the graft harvest were graded as A or B for 31 patients (83.8%), C for 3 patients (8.1%), and D for 3 patients (8.1%). Three patients with grade D had severe kneeling pain. In the 3 grade C patients, 1 had moderate kneeling pain without decrease of activity level, and the others complained of moderate anterior knee pain during activity. Of the 35 patients in the BHB group, the grade was A or B for 34 patients (97.1%) and C for 1 patient (2.9%). The patient with the grade C had moderate anterior knee pain, and 8 patients with grade B had mild numbness of calf without pain. Thus, complaints of anterior kneeling pain were less common in the BHB group than in the BTB group ($P = .056$, power = 0.91) (Figure 3).

Analyses of other categories (symptoms, compartment findings, radiograph findings, and functional test results) showed no significant differences between the groups. According to the IKDC final evaluation, although the overall score tended to be better in the BHB group, there were no significant differences between the groups ($P = .18$, power = 0.85).

Muscle Strength

A side-to-side ratio of the peak torque value at 60 and 180 deg/s was evaluated as a parameter (Table 5). A significant loss in isokinetic hamstring strength at 60 deg/s was observed in the BHB group ($P = .0045$, power = 0.91). In the other measures of isokinetic muscle strength, there were no significant differences between the 2 groups.

Postoperative Complications

There were no infections or graft ruptures, and no other major complications were encountered in either graft group. Twelve patients in the BTB group required additional operative procedures during the follow-up period. Among those, 8 patients (21.6%) developed meniscal symptoms and underwent partial meniscectomies (4 medial, 4 lateral), and 4 patients (10.8%) underwent notchplasty or excision of the cyclops lesion due to an extension deficit

at about 6 months postoperatively. On the other hand, 7 patients in the BHB group (20.0%) received partial meniscectomies (6 medial, 1 lateral) for subsequent meniscal injuries. Of these late meniscectomies, 4 patients in each group had previously undergone repair to their menisci.

DISCUSSION

The autogenous BTB graft and the hamstring tendon graft are the most commonly used in ACL reconstruction. There have been a number of studies comparing the clinical results of these grafts.¹¹ Although somewhat variable results have been reported, most of the previous studies have shown comparable results in both grafts, whereas some have shown a slightly increased incidence of residual laxity in the use of the hamstring tendon graft.^{11,24}

However, the majority of those studies compared the 2 types of grafts using different graft constructs and fixation methods. When the hamstring tendon graft was used, it was usually a multistrand, free-tendon graft fixed extra-articularly at a nonanatomical location; however, the BTB graft was fixed with interference screws close to the original insertion point. It has been shown that extra-articular fixation may result in unfavorable graft-bone tunnel motion and inferior efficacy in stabilizing the knee.^{14,19} Thus, the tendency toward inferior results in postoperative stability observed in the hamstring tendon graft may be attributed to this potential disadvantage of graft fixation method.

To our knowledge, there have been few studies comparing both grafts with comparable methods of fixation. Marder et al¹⁸ employed a post-and-washer fixation both proximally and distally, demonstrating no significant differences between the 2 grafts with respect to clinical results and objective stability. In a comparative study in which an interference-fit screw was used for the fixation of both grafts, Corry et al⁶ showed no significant differences in the results, whereas Shaieb et al²³ and Ejerhed et al⁷ observed more anterior knee pain in the patellar tendon group. In these studies, however, the hamstring tendon was used as a free-tendon graft, with no bony segment in the construct.

To overcome the theoretical disadvantages of free-tendon grafts in the use of the hamstring tendon graft, the idea of a BHB graft was initially proposed by Morgan et al¹⁹ and Liu et al.¹⁷ Novak et al²⁰ showed that a soft tissue interference-screw fixation with free-bone block has superior initial strength to a soft tissue fixation using sutures tied over a post. Weiler et al²⁶ demonstrated that a soft tissue graft fixation with an interference screw could increase the initial pullout force through the use of a bone plug. Moreover, the graft incorporation process in the bone tunnel is expected to be enhanced by the presence of the bone plug.

In this study, a comparison was made between the BTB and BHB grafts using the same construct and fixation methods. Thus, it was expected that the effect of graft selection on the outcome of ACL reconstruction could be critically evaluated. In the study, the mean KT-1000 arthrometer results at manual maximum stress showed no

significant difference between the BTB and BHB groups. The apparently improved stability result of the hamstring tendon ACL graft in this study compared with some of the previous reports may be attributable to the adoption of the BHB graft construct. On the other hand, when symptoms related to graft harvest were compared between the 2 groups, a distinct difference was delineated. In the comparison of graft harvest site morbidity, anterior kneeling pain was clearly lower in the BHB group than in the BTB group. In the 3 BTB patients with a final evaluation of grade D, unsatisfactory results were correlated with severe kneeling pain in all cases. On the other hand, graft harvest site morbidity in the BHB patients was mild numbness (grade B). Tadokoro et al²⁵ observed that one symptom related to graft harvest was a reduction in hamstring strength during deep knee flexion. However, this hamstring muscle weakness generally does not appear as a significant functional deficit in daily activities. We observed significant loss in isokinetic hamstring strength at 60 deg/s in the BHB group without complaint of functional deficit. Therefore, the hamstring tendon graft in ACL reconstruction appears to have an advantage over the patellar tendon graft in terms of reducing the risk of problems associated with graft harvest while maintaining comparable results in the remaining clinical parameters.

One issue in the present study is a difference between the 2 groups in gender distribution and length of follow-up periods, but there was no significant difference between the groups. Another issue is the high reoperation rate in the present study. Twenty-two percent of patients in the BTB group and 20% of those in the BHB group underwent partial meniscectomies after ACL reconstruction. After meniscal repair, 26.7% of patients in the BTB group and 28.6% of patients in the BHB group underwent later meniscectomy. In previous studies, the healing rate after meniscal repair was shown to be between 76% and 94%.^{4,5,16} Our results showed a slightly higher rate of repeat tear after meniscal repair.

Our study is the first to objectively compare patellar tendon and hamstring tendon autografts using the same graft construct and same fixation method. Reconstruction of the ACL using BTB grafts demonstrated more abnormal and severely abnormal results due to kneeling pain. On a practical level, however, there are some shortcomings in the use of BHB grafts, such as additional surgery time for graft preparation, technical complexity, and difficulty in its application to the 2-bundle anatomical reconstruction. Thus, the reconstructive procedure using the hamstring tendon graft requires further refinement, at the same time retaining the advantage of less harvest site morbidity. Conversely, in ACL reconstruction with the BTB graft, the possibility of preventive measures for reducing donor pain, such as modified rehabilitation programs, should also be pursued.

REFERENCES

1. Aglietti P, Buzzi R, Zaccherotti G, De Biase P. Patellar tendon versus doubled semitendinosus and gracilis tendons for anterior cruciate ligament reconstruction. *Am J Sports Med.* 1994;22:211-217, discussion 217-218.
2. Aune AK, Ekeland A, Cawley PW. Interference screw fixation of hamstring vs patellar tendon grafts for anterior cruciate ligament reconstruction. *Knee Surg Sports Traumatol Arthrosc.* 1998;6:99-102.

¹¹References 1-3, 6, 7, 9, 10, 15, 18, 21-23.

3. Beynon BD, Johnson RJ, Fleming BC, et al. Anterior cruciate ligament replacement: comparison of bone-patellar tendon-bone grafts with two-strand hamstring grafts. A prospective, randomized study. *J Bone Joint Surg Am*. 2002;84:1503-1513.
4. Buseck MS, Noyes FR. Arthroscopic evaluation of meniscal repairs after anterior cruciate ligament reconstruction and immediate motion. *Am J Sports Med*. 1991;19:489-494.
5. Cannon WD Jr. Arthroscopic meniscal repair: inside-out technique and results. *Am J Knee Surg*. 1996;9:137-143.
6. Corry IS, Webb JM, Clingeleffer AJ, Pinczewski LA. Arthroscopic reconstruction of the anterior cruciate ligament: a comparison of patellar tendon autograft and four-strand hamstring tendon autograft. *Am J Sports Med*. 1999;27:444-454.
7. Ejerhed L, Kartus J, Sernert N, Kohler K, Karlsson J. Patellar tendon or semitendinosus tendon autografts for anterior cruciate ligament reconstruction? A prospective randomized study with a two-year follow-up. *Am J Sports Med*. 2003;31:19-25.
8. Erdfelder E, Faul F, Buchner A. G*Power: a general power analysis program. *Behav Res Methods Instr Comput*. 1996;28:1-11.
9. Eriksson K, Anderberg P, Hamberg P, et al. A comparison of quadruple semitendinosus and patellar tendon grafts in reconstruction of the anterior cruciate ligament. *J Bone Joint Surg Br*. 2001;83:348-354.
10. Feller JA, Webster KE. A randomized comparison of patellar tendon and hamstring tendon anterior cruciate ligament reconstruction. *Am J Sports Med*. 2003;31:564-573.
11. Freedman KB, D'Amato MJ, Nedeff DD, Kaz A, Bach BR Jr. Arthroscopic anterior cruciate ligament reconstruction: a metaanalysis comparing patellar tendon and hamstring tendon autografts. *Am J Sports Med*. 2003;31:2-11.
12. Hefti F, Muller W, Jakob RP, Staubli HU. Evaluation of knee ligament injuries with the IKDC form. *Knee Surg Sports Traumatol Arthrosc*. 1993;1:226-234.
13. Irrgang JJ, Anderson AF, Boland AL, et al. Development and validation of the International Knee Documentation Committee subjective knee form. *Am J Sports Med*. 2001;29:600-613.
14. Ishibashi Y, Rudy TW, Livesay GA, Stone JD, Fu FH, Woo SL. The effect of anterior cruciate ligament graft fixation site at the tibia on knee stability: evaluation using a robotic testing system. *Arthroscopy*. 1997;13:177-182.
15. Jansson KA, Linko E, Sandelin J, Harilainen A. A prospective randomized study of patellar versus hamstring tendon autografts for anterior cruciate ligament reconstruction. *Am J Sports Med*. 2003;31:12-18.
16. Kurosaka M, Yoshiya S, Kuroda R, Matsui N, Yamamoto T, Tanaka J. Repeat tears of repaired menisci after arthroscopic confirmation of healing. *J Bone Joint Surg Br*. 2002;84:34-37.
17. Liu SH, Kabo JM, Osti L. Biomechanics of two types of bone-tendon-bone graft for ACL reconstruction. *J Bone Joint Surg Br*. 1995;77:232-235.
18. Marder RA, Raskind JR, Carroll M. Prospective evaluation of arthroscopically assisted anterior cruciate ligament reconstruction: patellar tendon versus semitendinosus and gracilis tendons. *Am J Sports Med*. 1991;19:478-484.
19. Morgan CD, Kalmam VR, Grawl DM. Isometry testing for anterior cruciate ligament reconstruction revisited. *Arthroscopy*. 1995;11:647-659.
20. Novak PJ, Wexler GM, Williams JS Jr, Bach BR Jr, Bush-Joseph CA. Comparison of screw post fixation and free bone block interference fixation for anterior cruciate ligament soft tissue grafts: biomechanical considerations. *Arthroscopy*. 1996;12:470-473.
21. O'Neill DB. Arthroscopically assisted reconstruction of the anterior cruciate ligament: a prospective randomized analysis of three techniques. *J Bone Joint Surg Am*. 1996;78:803-813.
22. Otero AL, Hutcheson L. A comparison of the doubled semitendinosus/gracilis and central third of the patellar tendon autografts in arthroscopic anterior cruciate ligament reconstruction. *Arthroscopy*. 1993;9:143-148.
23. Shaieb MD, Kan DM, Chang SK, Marumoto JM, Richardson AB. A prospective randomized comparison of patellar tendon versus semitendinosus and gracilis tendon autografts for anterior cruciate ligament reconstruction. *Am J Sports Med*. 2002;30:214-220.
24. Spindler KP, Kuhn JE, Freedman KB, Matthews CE, Dittus RS, Harrell FE Jr. Anterior cruciate ligament reconstruction autograft choice: bone-tendon-bone versus hamstring. Does it really matter? A systematic review. *Am J Sports Med*. 2004;32:1986-1995.
25. Tadokoro K, Matsui N, Yagi M, Kuroda R, Kurosaka M, Yoshiya S. Evaluation of hamstring strength and tendon regrowth after harvesting for anterior cruciate ligament reconstruction. *Am J Sports Med*. 2004;32:1644-1650.
26. Weiler A, Hoffmann RF, Stahelin AC, Bail HJ, Siepe CJ, Sudkamp NP. Hamstring tendon fixation using interference screws: a biomechanical study in calf tibial bone. *Arthroscopy*. 1998;14:29-37.